

Cross-Cultural Dynamics of Emotional Expression in Lyrics

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Website: <https://com-480-data-visualization.github.io/lyrical-emotion-InSight/>

We present two complementary parts to study and visualize emotion expression in lyrics: (1) a cross-cultural, diachronic analysis of 1.2 million songs (1950–2020) in 15 languages—annotated with transformer-derived VAD (i.e. valence, arousal, dominance) emotional space provided by the WASABI dataset noted in milestone 1—to compute static emotion distributions and dynamic sentiment trends via regression and change-point detection; and (2) exploring the relationship between the narrative in lyrics and emotion episodes, based on the Episode Model [2] that maps linguistic structures (e.g., repetition, metaphor, valence arcs) to functional emotion episodes, revealing both universal and culture-specific patterns of lyrical emotion.

Part 1: Emotional Expression in Lyrics

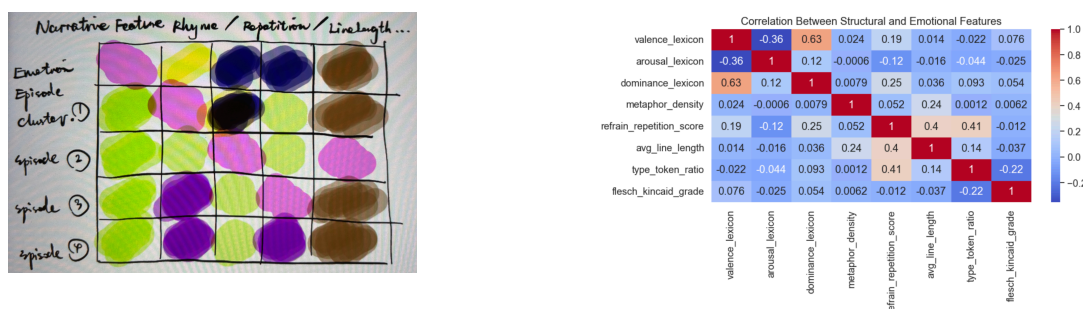
Static Analysis: aggregate distribution of the VAD emotional space by language to identify significant cross-linguistic differences. **Dynamic Analysis:** 1) time-series of positive vs. negative word usage in each language, visualized via moving averages (decade windows); 2) linear regression of emotion scores over time, yielding per-language slope coefficients and R^2 values; interactive scatterplots with fitted lines and data-point tooltips enable drill-down to language-level trend demonstration.

These analyses are accessible through our prototype web interface, which supports filtering by language and emotion dimension. We'll also analyze country-level effects on lyrical emotion—comparing static distributions, dynamic trends to uncover how geography shapes emotional expression in song lyrics.

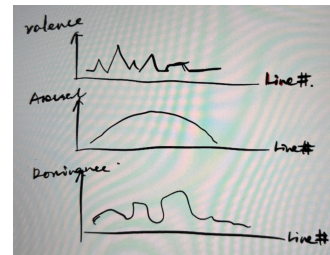
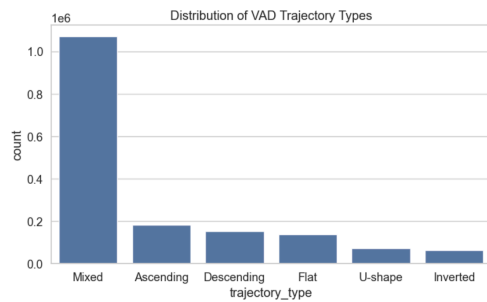
Part 2: Emotional Expression in Lyrics

We then examine how lexical and structural cues shape lyrical emotion by first annotating each lyric line with valence–arousal–dominance (VAD) scores derived from the NRC Emotion Lexicon [1]. We extract a suite of interpretable features—repetition frequency, metaphor density (via metaphor identification algorithm), and narrative progression (measured by semantic shift across successive lines)—and statistically relate these features to functional emotion episodes as defined by the Episode Model [2].

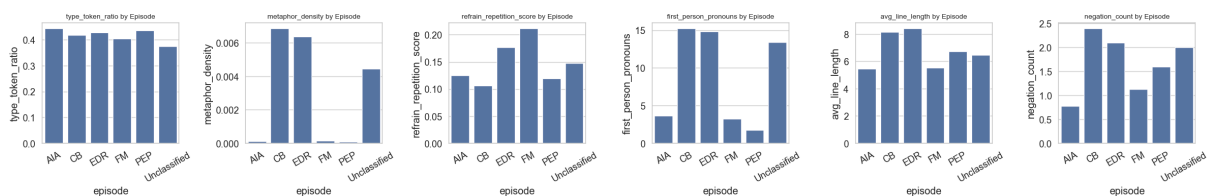
Heatmap of Emotional Episode Clusters and Structural Feature Correlations



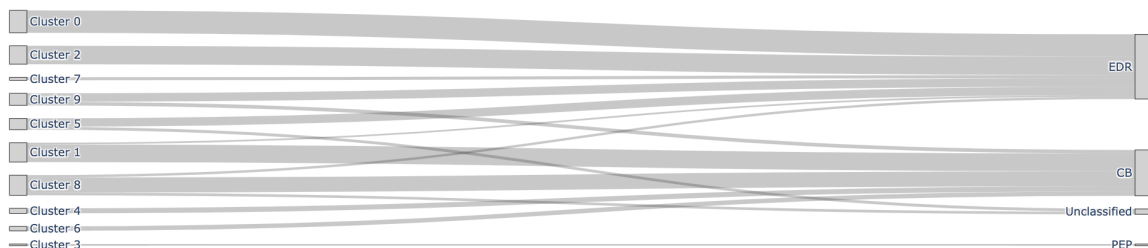
Song-Level Emotion Trajectories & Overall Shape Distribution



Linking Narrative Structure to Emotional Episode Trajectories



🔗 KMeans Cluster on VAD → Episode Mapping



Tools & Lecture Dependencies

Visualization	Python Tooling	Lecture
VAD distribution plots	pandas + seaborn.kdeplot / histplot	11.1 Tabular Data
VAD trajectory (line-by-line)	matplotlib.pyplot.plot	11.1 Tabular Data
UMAP scatter of affective embeddings	umap-learn, scikit-learn (KMeans), matplotlib	11.1 Tabular Data / 7.1 Designing Viz
Episode-cluster bar chart	seaborn.barplot	11.1 Tabular Data
Structural-feature heatmap	seaborn.heatmap	11.1 Tabular Data
Interactive dashboard (filtering, linking)	plotly Dash or streamlit	5.1 Interaction Methods
Word-clouds per episode (optional)	wordcloud	9 Text Visualization
Network graph of structural motifs (optional)	networkx + matplotlib or pyvis	10 Graph Visualization

Extra ideas: the relationship between musical features (chord progression, scale, tonality) with the emotional content of lyrics is also an interesting direction to explore.

References

- [1] Mohammad, S. M., & Turney, P. D. (2013). Nrc emotion lexicon. *National Research Council, Canada*, 2, 234.
- [2] Tuomas Eerola, Christina Kirts, and Suvi Saarikallio. 2024. Episode model: The functional approach to emotional experiences of music. *Psychology of Music*.